

Scalability & Future Plan

Date	06 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Currently supports only text-based emotion detection.	Cannot analyze emotions from voice or facial expressions.	High
2	Interaction history is stored in CSV files.	Limited scalability and data management for large numbers of users.	High
3	AI response depends on internet connectivity and Gemini API availability.	Response time may increase during network issues or API delays.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users on Streamlit Community Cloud.	Deploy on cloud platforms (AWS, Azure, or GCP) with load balancing to support more concurrent users.
2	Data Storage	Uses CSV files for interaction history and analytics.	Replace CSV storage with SQLite , PostgreSQL , or MongoDB for efficient and scalable data management.
3	Performance	Emotion detection uses BiLSTM, BERT, and Gemini sequentially.	Optimize model inference, cache repeated requests, and use GPU acceleration where available.
4	Security	Gemini API key is securely configured for the application.	Add user authentication, role-based access control, encrypted data storage, and secure API management.

Future Plan

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add multilingual emotion detection and personalized guidance.	Next 3 Months	Support students from different language backgrounds.
Phase 2	Integrate voice-based emotion detection using speech recognition.	Next 6 Months	Enable hands-free interaction and improve accessibility.
Phase 3	Implement facial emotion recognition alongside text analysis.	Next 9 Months	Improve prediction accuracy through multimodal emotion detection.
Phase 4	Develop mobile applications and integrate with Learning Management Systems (LMS).	Next 12 Months	Increase accessibility and simplify deployment in educational institutions.